

Oxygen and Carbon Dioxide Transport

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 Adapted from: [Oxygen and Carbon Dioxide Transport](#)

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Learning Objectives (4)

Versions

After completing this brick, you will be able to:

- 1 Describe the mechanisms by which oxygen (O_2) and carbon dioxide (CO_2) are transported in the blood.
- 2 Explain the importance of the blood CO_2 concentration.
- 3 Relate the oxygen-hemoglobin dissociation curve to gas transport.
- 4 Apply the Haldane and Bohr effects to model oxygen and CO_2 uptake and release at the lung and peripheral tissues.

**CASE CONNECTION**

MM, a 50-year-old woman, presents with fatigue and exertional dyspnea. She reports heavy menstrual periods. Despite normal findings on pulmonary and cardiac examinations, her complete blood count reveals low hemoglobin concentration.

How can anemia lead to symptoms of tissue hypoxia despite normal respiratory and cardiac function? To answer this, we must explore how oxygen and carbon dioxide are transported in the blood—and how alterations in this system can impair oxygen delivery to tissues.

GO TO CONCLUSION ↓

Introduction

The primary function of the circulatory system is to transport respiratory gases: delivering O_2 from the lungs to peripheral tissues and removing CO_2 , a byproduct of metabolism, for exhalation via the lungs. Since small amounts of these gases dissolve in plasma, the movement of gases therefore depends largely on hemoglobin reversibly binding O_2 and the carbonic anhydrase enzyme in the RBC converting CO_2 into bicarbonate ions (HCO_3^-).

Oxygen transport in the blood (LO1)

Oxygen is transported in blood through two principle forms:

- Hemoglobin within red blood cells (98%) to form oxyhemoglobin. Hemoglobin's capacity to reversibly bind oxygen molecules makes it indispensable for efficient transport.
- Dissolved in plasma (2%). This is low because oxygen is poorly soluble in plasma

Hemoglobin consists of four globin subunits, each containing a heme moiety with a central iron atom capable of reversibly binding one oxygen molecule to form oxyhemoglobin (Figure 1). Thus, a single hemoglobin molecule can carry up to four oxygen molecules. Crucially, hemoglobin exhibits **cooperative binding**: the binding of one oxygen molecule increases the affinity of the remaining sites. This property underlies the characteristic sigmoid shape of the oxygen-hemoglobin dissociation curve (Figure 4).

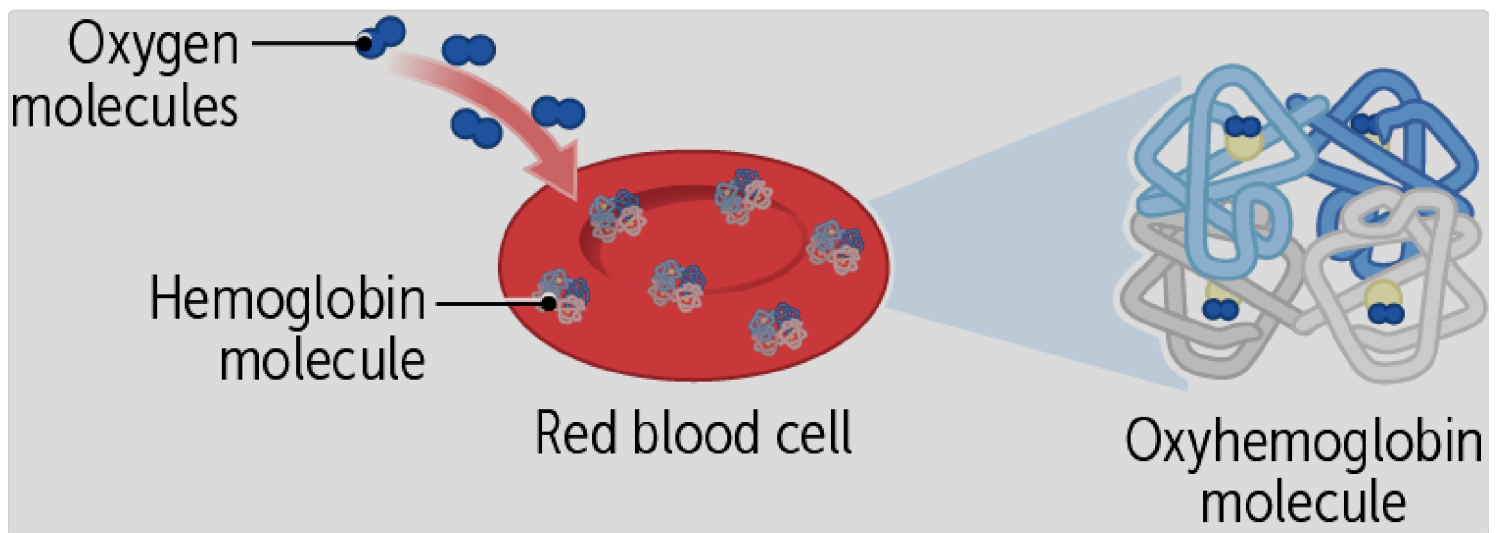


Figure 1 Oxygen binding to hemoglobin to form oxyhemoglobin

Measures of Oxygen in the Blood

To fully understand oxygen transport, it is important to note that oxygen in the blood is described by three distinct measures, partial pressure, saturation, and content, each capturing a different aspect of oxygen transport. Distinguishing between them is critical:

- **Partial pressure of oxygen (P_aO_2):** is the **free, unbound oxygen molecules dissolved directly in plasma**. Although only about 2% of oxygen is transported dissolved in plasma, the partial pressure of oxygen (P_aO_2) represents the driving force for oxygen diffusion from the alveoli into the deoxygenated blood (~100 mmHg vs ~40 mmHg). This dissolved oxygen then diffuses into red blood cells and binds to hemoglobin.
- **Oxygen saturation (S_aO_2 or SpO_2):** is the term used to describe the **percentage of hemoglobin's oxygen-binding sites that are occupied by oxygen**. It is calculated as $(\text{oxygenated Hb} \div \text{total Hb}) \times 100$. Each hemoglobin molecule has four binding sites, so if three of four site are occupied, saturation is 75% (normal ~98%). Importantly, because this is a **ratio**, it can mask dangerously reduced oxygen delivery to tissues with low hemoglobin levels in anemia. S_aO_2 is measured directly via arterial blood gas, while SpO_2 is estimated non-invasively by pulse oximetry.
- **Oxygen content (C_aO_2):** is the **total amount of oxygen** and depends on hemoglobin concentration, hemoglobin saturation, and P_aO_2 . Unlike saturation or P_aO_2 , C_aO_2 reflects the true oxygen-carrying capacity of blood. Normal arterial C_aO_2 is about 20 mL/dL, while mixed venous C_aO_2 at rest is approximately 15 mL/dL and decreases further during exercise. Importantly, oxygen bound to hemoglobin does not contribute to partial pressure, so changes in oxygen content or saturation do not directly affect the diffusion gradient.

The oxygen content is described below, reflecting the roles of dissolved oxygen, and HB loading and % saturation.

Where:

- **1.36 mL of oxygen per gram of hemoglobin:** is the oxygen-carrying capacity of hemoglobin (slightly higher in fetal hemoglobin, slightly lower in hemoglobin S)
- **[Hb]:** is the hemoglobin concentration in g/dL, measured as part of a complete blood count (CBC)
- **S_aO₂:** Arterial oxygen saturation (as a percent), measured on a pulse oximeter, and is expressed as a decimal
- **Solubility coefficient of oxygen** in plasma: 0.003 mL O₂/mmHg/dL (which is low)
- **P_aO₂:** Arterial partial pressure of oxygen, measured on an arterial blood gas



CLINICAL REASONING

Anemia reduces oxygen delivery by lowering hemoglobin concentration, even when lung function is normal. Patients may develop exertional hypoxia with fatigue, dyspnea, and pallor. The problem is best detected by measuring **arterial oxygen content (CaO₂)**, which reflects the total amount of hemoglobin-bound oxygen in blood. In contrast, **oxygen saturation (SpO₂ or SaO₂)** remains normal because it measures the percentage of hemoglobin binding sites occupied, not the amount of hemoglobin present. **PaO₂** is also normal because dissolved oxygen in plasma is unchanged.

For example, a patient with severe anemia (Hgb 7 g/dL, normal Hb ~ 15 mg/dL) may have a normal saturation of 98% but a markedly reduced oxygen content (~9.6 mL/dL vs. ~20 mL/dL normally). Thus, normal saturation can mask impaired oxygen delivery, making hemoglobin concentration essential when evaluating tissue oxygenation.

with only a small fraction carried dissolved in plasma. capacity. **98% of oxygen is transported bound to hemoglobin**, binds oxygen, greatly increasing the blood's oxygen-carrying capacity. Hemoglobin within red blood cells transports oxygen in dissolved form. Hemoglobin within red blood cells transports oxygen in dissolved form. Hemoglobin within red blood cells transports oxygen in dissolved form. Oxygen is poorly soluble in plasma, so only a small amount can be

Carbon Dioxide Transported in Blood (LO1)

As tissues carry out metabolism, they produce carbon dioxide (CO_2) as a waste product. When dissolved in blood, CO_2 forms carbonic acid, which can disrupt normal cellular function if it accumulates. Therefore, CO_2 must be efficiently transported from tissues to the lungs for exhalation. This transport occurs primarily in the plasma, with a smaller portion carried

within red blood cells. Carbon dioxide is transported in blood in **three primary** forms:

- Bicarbonate ions in plasma (~85%) - most important form
- Bound to hemoglobin (~10%)
- Dissolved in plasma (~5%)

CO₂ transported as Bicarbonate (HCO₃⁻)

CO₂ conversion to bicarbonate is catalyzed by carbonic anhydrase within red blood cells, and the direction of the reaction reverses depending on location — forming bicarbonate in the tissues and regenerating CO₂ at the lungs. *High yield point:* Remember that carbonic anhydrase catalyzes its reaction in both directions with the same rate, meaning that the concentration of reactants drives the shift of CO₂ to bicarbonate and back again.

In peripheral tissues, CO₂ diffuses into red blood cells and combines with water to form carbonic acid (H₂CO₃), catalyzed by carbonic anhydrase. This unstable intermediate rapidly dissociates into H⁺ and HCO₃⁻. The H⁺ binds to hemoglobin, which acts as a buffer. The HCO₃⁻ is transported out into the plasma.

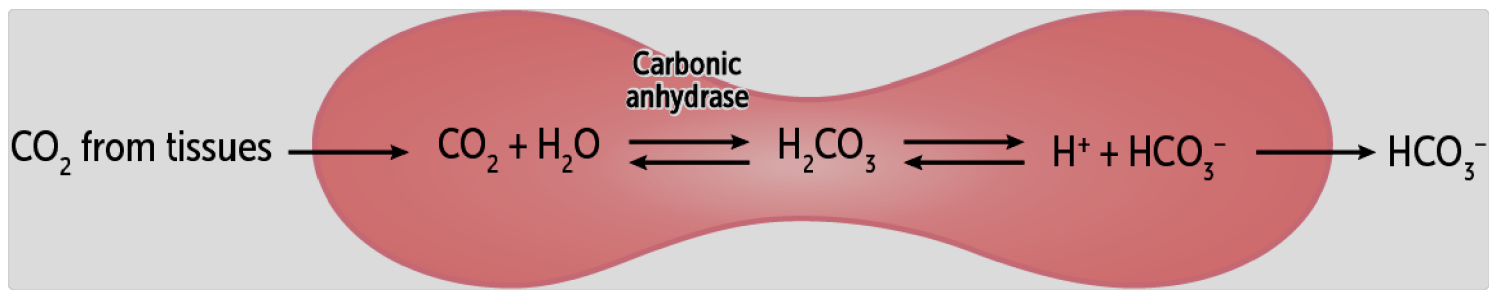
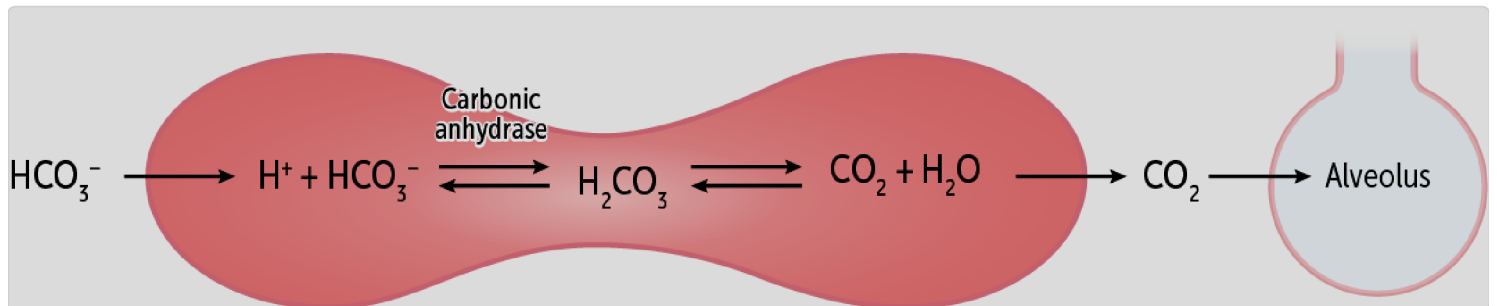


Figure 2 Bicarbonate formation in RBCs

At the lungs, the process reverses. HCO_3^- re-enters the red blood cells in exchange for chloride. Inside the cell, hemoglobin releases its buffered H^+ , which recombines with HCO_3^- . Carbonic anhydrase catalyzes the formation of H_2CO_3 , which dissociates into CO_2 and water. The CO_2 then diffuses into the plasma, crosses into the alveolus, and is exhaled.

Figure 3 Release of CO₂ from RBCs at the lung

CO₂ Bound to Hemoglobin

A smaller fraction of CO_2 binds directly to the amino groups of hemoglobin, forming **carbaminohemoglobin** (10%). Deoxygenated hemoglobin is more effective at carrying CO_2 from tissues to lungs. The erythrocytes carry carbaminohemoglobin to the lungs where the concentration of CO_2 is

relatively low. In the lungs, CO_2 is released from the hemoglobin and into the alveoli for excretion.

CO_2 Dissolved in Plasma

The remaining 5% of CO_2 is transported as dissolved gas directly in plasma. Recall that about 2% of oxygen is dissolved in plasma, so CO_2 is about 20% more soluble than oxygen. This accounts for the higher (if still low) percentage of CO_2 dissolved in plasma.

RBCs accelerates the conversion.

82% of CO_2 is transported as bicarbonate. Carbonic anhydrase in

Importance of Blood Carbon Dioxide Concentration (LO2)

Role of Carbon Dioxide in pH Regulation

Carbon dioxide plays a key role in regulating blood pH through the reversible reactions catalyzed by carbonic anhydrase in red blood cells as shown in [Figures 2](#) and [Figure 3](#). Blood pH is determined by the $[\text{HCO}_3^-]/[\text{CO}_2]$ ratio which represents the chemical equilibrium for this buffer system (simplified from the Henderson-Hasselbalch equation). Furthermore, total $[\text{HCO}_3^-]$ and $[\text{CO}_2]$ are regulated independently of each other in the body allowing pH adjustments via quick changes at the lungs (changing the rate of CO_2 removal), or metabolic changes at the kidneys (renal HCO_3^- reabsorption, secretion, and production), depending on the needs of the system.

Increased CO_2 , as in [hypoventilation](#) or [respiratory failure](#), drives the catalytic reaction toward the production of increased bicarbonate and H^+ ([Figure 2](#)), and lowering pH ([respiratory acidosis](#)). Conversely, [hyperventilation](#) reduces CO_2 and $[\text{H}^+]$, raising pH ([respiratory alkalosis](#)). Hemoglobin buffers much of the liberated H^+ , especially when deoxygenated, linking respiratory gas transport to acid-base regulation.

Also, hypercapnia can cause dyspnea and vasodilation.

which must be carefully maintained for normal cell function.

Blood CO_2 concentration is an important determinant of blood pH,

 CLINICAL CORRELATION**CO₂ and Acid-Base Disorders**

Disruption of CO₂ handling leads to acid-base disorders with significant clinical implications. For example, in **chronic obstructive pulmonary disease (COPD)**, impaired ventilation leads to CO₂ retention and respiratory acidosis.

Renal compensation occurs slowly by increasing H⁺ excretion and bicarbonate reabsorption.

In contrast, hyperventilation during **anxiety or high-altitude exposure** causes respiratory alkalosis, presenting with lightheadedness, paresthesias, and even syncope due to cerebral vasoconstriction. Understanding how CO₂ governs pH is essential for interpreting arterial blood gases and managing critically ill patients.

Oxygen–Hemoglobin Dissociation Curve (LO3)

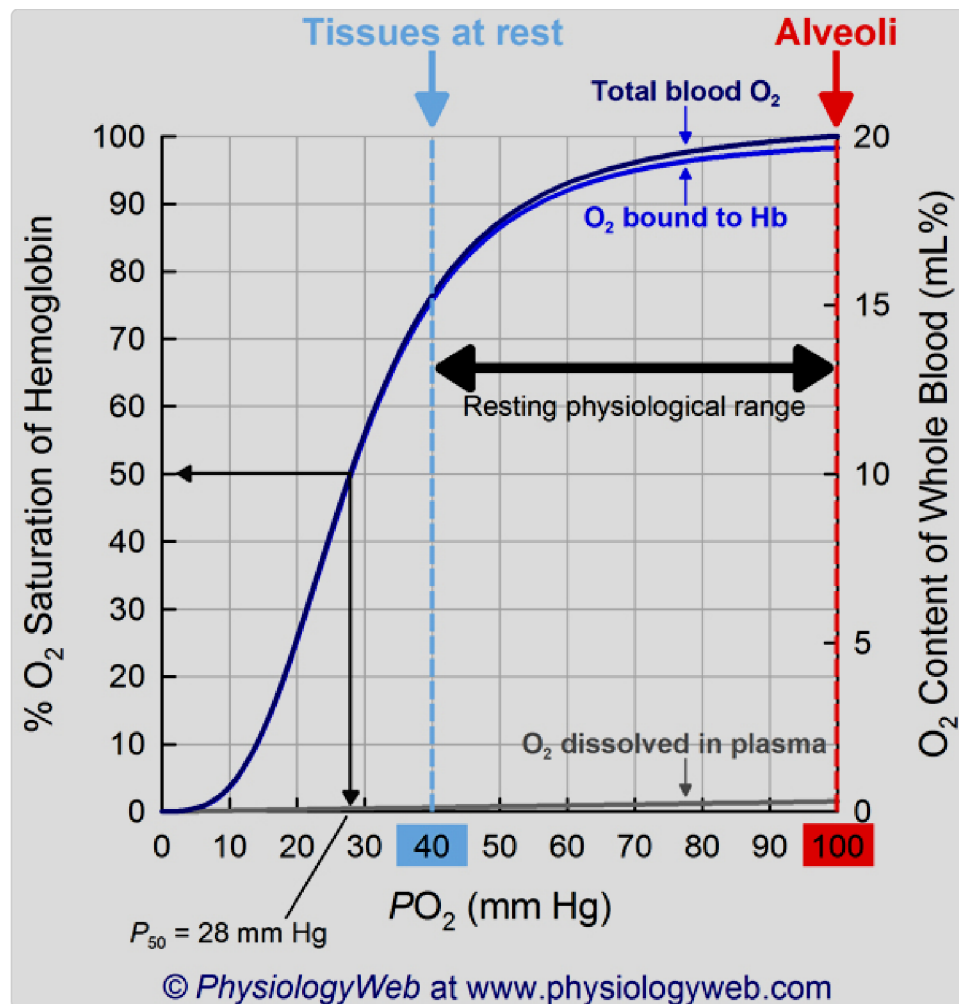


Figure 4. Oxygen-hemoglobin dissociation curve with 3 measures of O_2 (partial pressure, saturation and content)

The oxygen-hemoglobin dissociation curve plots PaO_2 on the x-axis and SaO_2 , and CaO_2 on the y-axes (Figure 4). Its sigmoid shape reflects cooperative binding, with two distinct regions: the plateau and the steep slope.

CLINICAL REASONING

In the **plateau phase** (P_aO_2 80–100 mmHg) oxygen is loaded onto hemoglobin in the lungs resulting in high oxygen saturation (>95%) and C_aO_2 of ~20 mL/dL. This plateau thus provides a buffer against any drops in inspired oxygen.

In the **steep portion of the curve** (P_aO_2 ~10–40 mmHg), oxygen is unloaded where it is needed - at the metabolically active tissues. Small decreases in PO_2 produce large decreases in hemoglobin saturation due to cooperative dissociation, with oxygen content falling from about 15 mL/dL to as low as 2.5–10 mL/dL.

At a PO_2 of ~40 mmHg, hemoglobin is ~75% saturated (C_aO_2 of 15 mL/dL), released only about 25% of its oxygen. During exercise, as tissue PO_2 plummets to 20 mmHg, an additional 50% of the oxygen reserve is released (30% sat; C_aO_2 of 10 mL/dL) to match the heightened metabolic demand saturation drops to 30% (), releasing.

Role of Bohr and Haldane effects in O_2 and CO_2 transport (LO4)

So far, oxygen and carbon dioxide uptake and release by red blood cells have been described separately, but in reality, oxygen and CO_2 interact to affect each other's binding to hemoglobin. The dissociation curve shifts dynamically in response to local conditions at both the lungs and tissues. The changes to CO_2 and O_2 transport are mediated by the **Bohr** and **Haldane** effects. CO_2 and H^+ affect hemoglobin's affinity for oxygen (Bohr effect), while O_2 drives hemoglobin's capacity to bind CO_2 and H^+ (Haldane effect).

The Bohr and Haldane effects are often taught separately, but they operate simultaneously in both the lungs and tissues to facilitate oxygen loading in

the lungs and unloading in the tissues with the removal of metabolic CO_2 from the body. Both processes arise from the binding of O_2 , CO_2 , and H^+ to hemoglobin. Hemoglobin acts as a molecular switch (Figure 5), promoting oxygen loading in the lungs and oxygen unloading in peripheral tissues while simultaneously facilitating carbon dioxide transport at these locations.

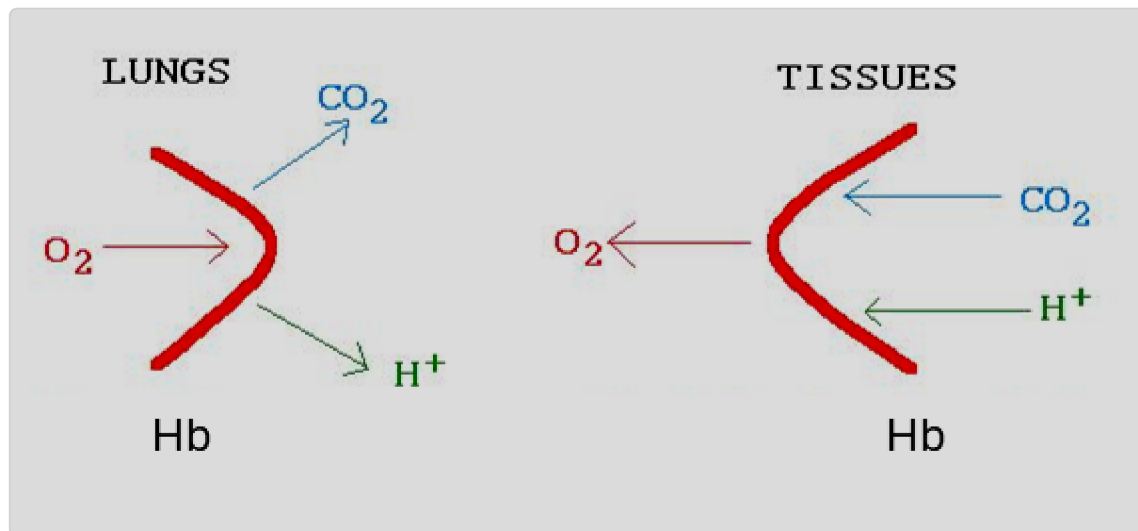


Figure 5. Hemoglobin acts as a molecular switch that couples oxygen delivery and carbon dioxide removal.

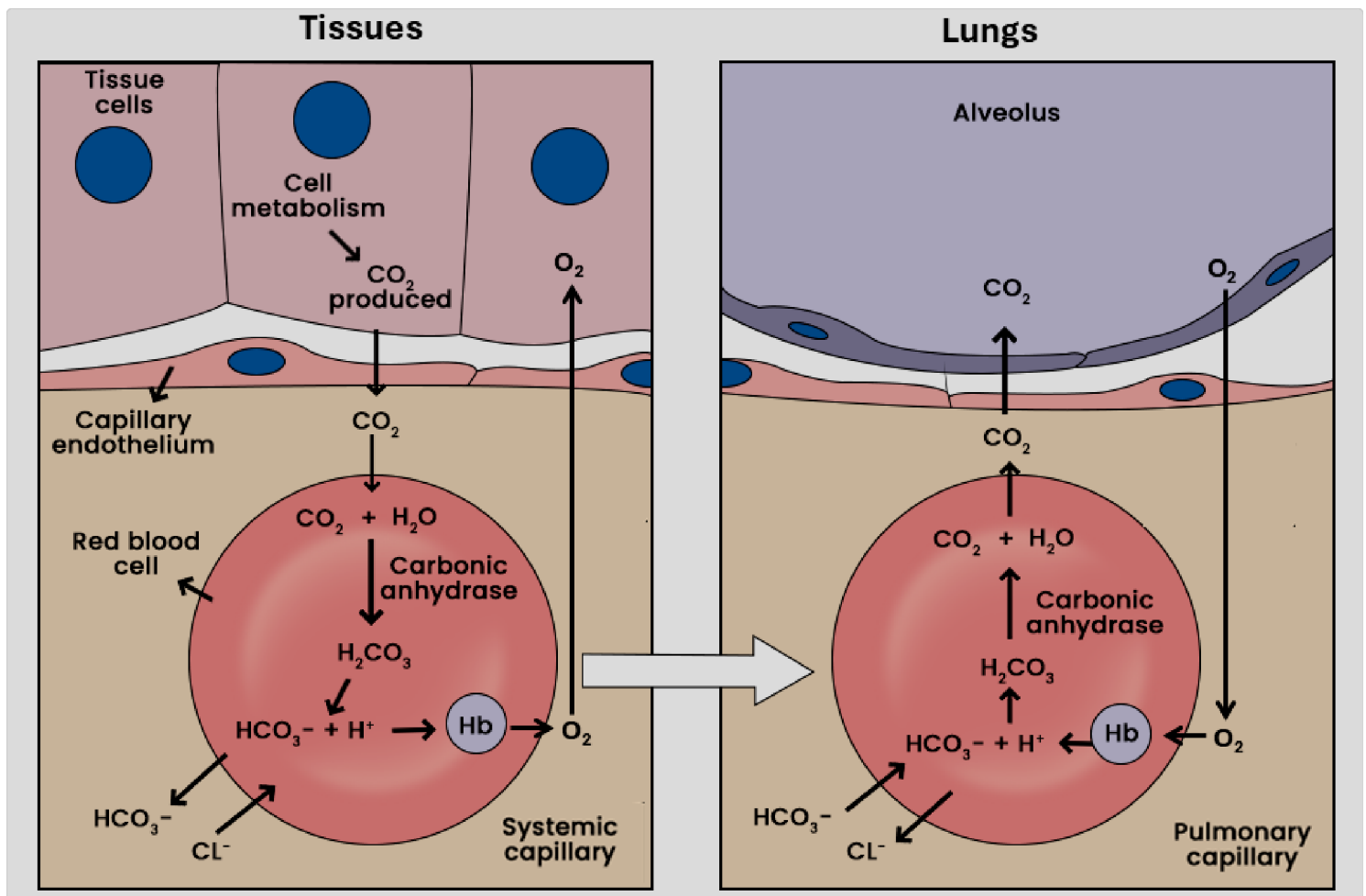


Figure 6. Low oxygen and H^+ concentrations in the tissue promote oxygen release and CO_2 binding to hemoglobin, while conditions in the lung promote oxygen binding and CO_2 release.

- **At the tissues (high CO_2 , low O_2)**

Metabolically active tissues produce carbon dioxide and hydrogen ions as by-products of cellular respiration. The rise in CO_2 and H^+ decreases hemoglobin's affinity for oxygen, (Bohr effect) (Figure 5, and Figure 6 - tissues). As a result, hemoglobin releases oxygen more readily, promoting oxygen unloading to the tissues where it is needed. At the same time, the decrease in oxygen saturation increases hemoglobin's capacity to bind carbon dioxide and hydrogen ions (Haldane effect) (Figure 5, and Figure 6 - tissues). The buffering of H^+ by

Hb drives the conversion of CO_2 to bicarbonate within red blood cells. Together, these processes allow efficient oxygen delivery to metabolically active tissues while facilitating uptake and transport of carbon dioxide in the blood.

- **At the lungs (high O_2 , low CO_2)**

In the lungs, ventilation removes carbon dioxide and increases the partial pressure of oxygen. The decrease in CO_2 and H^+ increases hemoglobin's affinity for oxygen promoting oxygen loading onto hemoglobin (Figure 5, and Figure 6 - lungs). Oxygen binding also releases carbon dioxide and hydrogen ions (Figure 5, and Figure 6 - lungs). The released H^+ recombines with bicarbonate to form carbonic acid, which is rapidly converted to CO_2 and water by carbonic anhydrase. The CO_2 then diffuses into the alveoli and is exhaled. These coordinated mechanisms allow efficient oxygen uptake in the lungs while promoting removal of carbon dioxide from the blood.

O_2 release in tissues (Boqyl).

H₂O + H⁺ = H₂O + $\text{O}_2 \rightarrow \text{CO}_2$ release in lungs. **BOyl** = H₂O + $\text{CO}_2 \rightarrow \text{H}^+$

conformational change that decreases its affinity for oxygen.
 H^+ binds to specific amino acids on hemoglobin, which causes a

The beauty of this system lies in its dynamic responsiveness: the curve doesn't remain static but shifts continuously as blood circulates, always positioning itself to optimize gas exchange for the current metabolic environment.

The **P₅₀ value** is the PO₂ at which hemoglobin is 50% saturated and quantifies hemoglobin's oxygen affinity. The normal P₅₀ is approximately 28 mmHg, placing it in the middle of the steep portion where oxygen unloading is most responsive to tissue needs. A **right shift** of the curve indicates decreased oxygen affinity, supports oxygen unloading in the tissues (Figure 7, left panel). When the **curve shifts left**, oxygen affinity is increased, supporting oxygen loading in the lungs (Figure 7, right panel). Clinically, P₅₀ provides a quantifiable metric for assessing how disease states, environmental conditions, or genetic variants alter hemoglobin's oxygen-binding behavior.

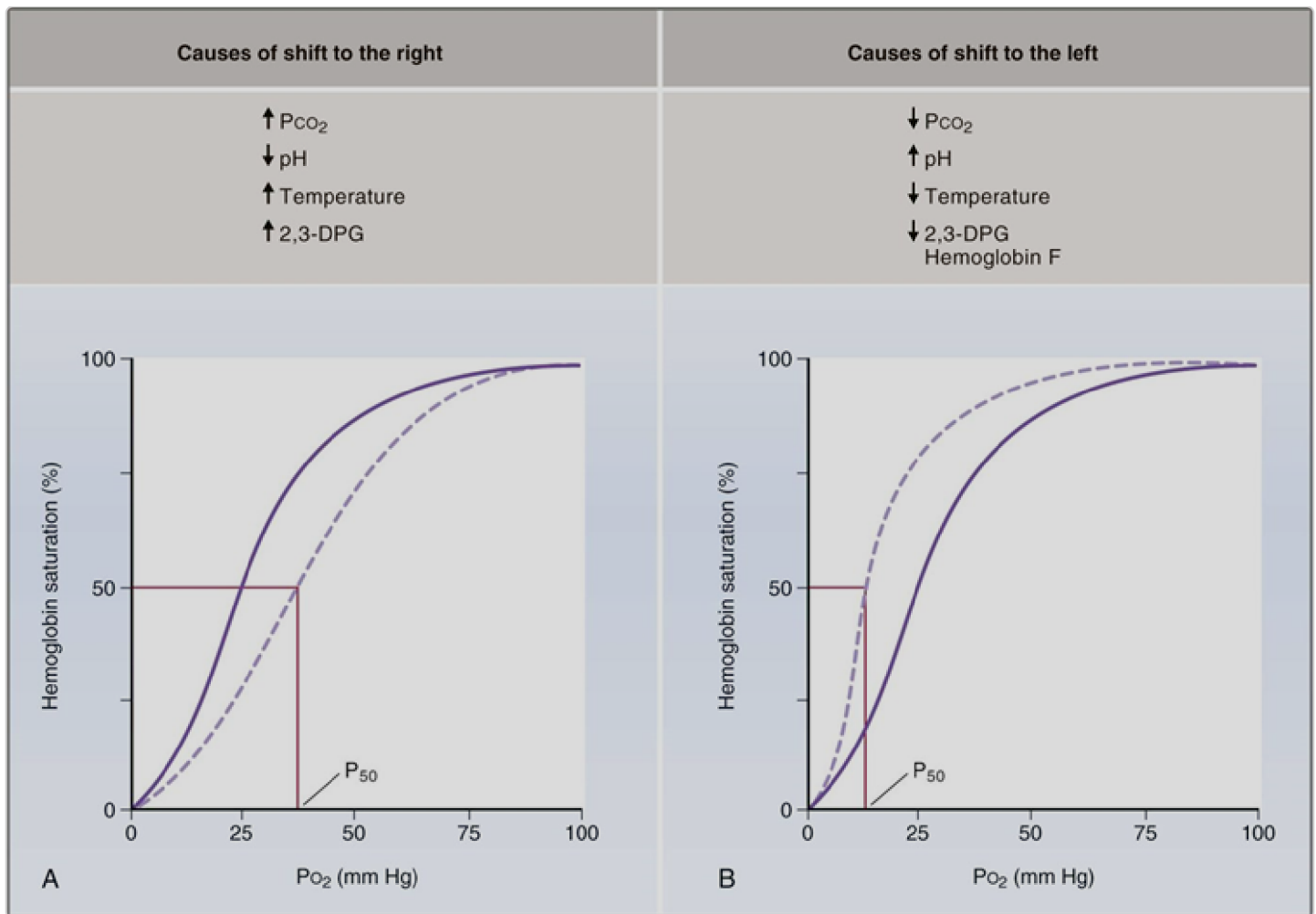


Figure 7. Shifting of the oxygen-hemoglobin dissociation curve to the left and right due to factors altering Hb affinity for oxygen

CLINICAL CORRELATION

Oxygen release: Right shift

Exercise: Working muscles generate heat, CO_2 , and lactic acid — a triple stimulus that shifts the curve right. Elevated temperature, increased PCO_2 , and decreased pH together maximize oxygen extraction precisely where metabolic demand is highest.

Septic shock: Tissue hypoxia drives CO_2 and lactate accumulation, lowering local pH. This acidosis enhances oxygen unloading via the Bohr effect, while

increased 2,3-BPG production from anaerobic metabolism reinforces the rightward shift — a crucial compensatory mechanism.

Chronic hypoxemia (high altitude or chronic lung disease): Sustained low PaO_2 stimulates erythrocyte 2,3-BPG production over days to weeks, improving oxygen release at the tissue level. This adaptation partially compensates for reduced arterial oxygen content and explains why chronically hypoxemic patients may tolerate surprisingly low saturations without acute symptoms.

Diabetic ketoacidosis: Severe metabolic acidosis from ketone accumulation shifts the curve right, facilitating oxygen delivery despite potential concurrent anemia and hypovolemia. However, this benefit diminishes below pH 7.2, where enzymatic dysfunction and other pathophysiologic effects outweigh improved oxygen unloading.



CLINICAL CORRELATION

Oxygen binding — Left shift

Fetal hemoglobin (HbF) has a left-shifted curve because its γ chains bind 2,3-DPG poorly, resulting in higher oxygen affinity. This allows HbF to out-compete maternal hemoglobin for oxygen at the placenta, facilitating efficient maternal-to-fetal transfer.

Anemia reduces total hemoglobin, diminishing oxygen content (C_aO_2) even when SaO_2 and PaO_2 remain normal. This is the basis of MM's symptoms: despite normal saturation, tissue oxygen delivery is compromised.

Carbon monoxide poisoning: CO binds hemoglobin with far greater affinity than oxygen, occupying binding sites and preventing O_2 transport. CO binding also increases the remaining sites' affinity for O_2 , shifting the curve left and

impairing tissue unloading. Pulse oximetry is falsely reassuring, as it cannot differentiate oxyhemoglobin from carboxyhemoglobin.

Methemoglobinemia: Oxidation of hemoglobin iron from ferrous (Fe^{2+}) to ferric (Fe^{3+}) prevents oxygen binding at affected sites. The remaining functional sites exhibit increased O_2 affinity (left shift), impairing release. Cyanosis with chocolate-colored blood is characteristic.

High-altitude adaptation: Chronic hypoxemia stimulates 2,3-BPG production, shifting the curve right and facilitating oxygen unloading at tissues.

Together, the Bohr and Haldane effects work simultaneously to ensure that hemoglobin automatically adjusts gas binding to match the local conditions in the body coordinate oxygen and carbon dioxide exchange through changes in hemoglobin binding of the ligands H^+ , CO_2 and O_2 (Figure 8).

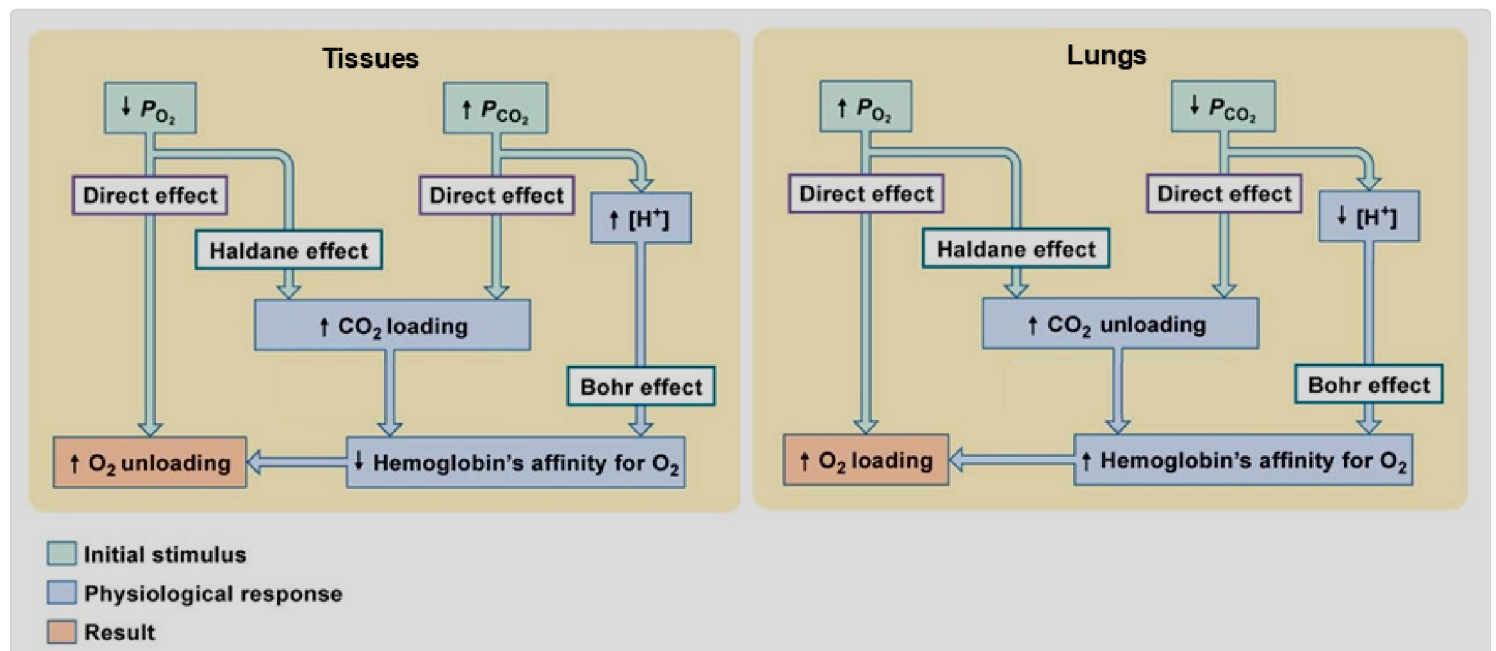


Figure 8. The interplay of H^+ , P_{CO_2} and P_{O_2} on O_2 and CO_2 unloading and loading

$E \rightarrow \downarrow \text{Exercise}$ ($\downarrow$ metabolic demand); and $T \rightarrow \downarrow \text{Temperature}$
 $2,3\text{-DPG}$ (released from RBC during hypoxia and increased alkalosis):
 $C \rightarrow \downarrow \text{CO}_2$ (hyperventilation); $A \rightarrow \downarrow \text{Acid}$ ($\uparrow$ pH, Bohr effect); $D \rightarrow \downarrow$



CASE CONNECTION

[BACK TO INTRODUCTION](#) 

Thinking back to MM, what CBC finding might relate to shortness of breath?

Returning to MM's case, her low hemoglobin level translates into a reduced oxygen-carrying capacity. While pulse oximetry may show a normal saturation, her oxygen content is significantly reduced. This diminished CaO_2 compromises tissue oxygen delivery, explaining her fatigue and dyspnea. Her case illustrates the critical distinction between saturation and content, underscoring the necessity of hemoglobin in effective oxygen transport.

Summary

- Oxygen is primarily transported in blood bound to hemoglobin in RBCs; a small amount is dissolved in plasma.

- The blood oxygen concentration (C_aO_2) can be calculated using the measured quantities of blood hemoglobin concentration (Hgb), oxygen saturation percentage (S_aO_2), and the arterial partial pressure of oxygen (P_aO_2).
- CO_2 is transported in the blood in 3 forms: the majority as plasma bicarbonate, bound to hemoglobin, and dissolved in the plasma.
- Plasma bicarbonate formation from CO_2 interaction with H_2O within the RBCs is accelerated by the enzyme carbonic anhydrase.
- The lungs and kidneys collaborate to keep the pH in a narrow range by excreting the CO_2 in the lungs and by generating bicarbonate in the kidneys.
- The Bohr and Haldane effects match hemoglobin binding of oxygen, CO_2 , and H^+ to the local conditions of the body as blood moves between the lungs and tissues.

Review Questions

1. AJ is a new intern on the medical wards. Her team admitted a patient experiencing significant respiratory difficulty. The senior resident asks her to calculate the patient's blood oxygen concentration as precisely as she can. Which of the following blood tests will best solve AJ's assignment?

- A. Arterial blood gas
- B. Arterial blood gas, pulse oximetry
- C. Arterial blood gas, pulse oximetry, blood hemoglobin level

- D. Arterial blood gas, pulse oximetry, blood hemoglobin level, blood mean hemoglobin concentration per RBC
- E. Pulse oximetry, blood hemoglobin level

Explanation (requires correct answer)

2. Which of the following blood conditions would cause hemoglobin to bind most avidly to oxygen?

- A. High $p\text{CO}_2$, high $[\text{H}^+]$
- B. High $p\text{CO}_2$, low $[\text{H}^+]$
- C. Low $p\text{CO}_2$, high $[\text{H}^+]$

D. Low $p\text{CO}_2$, low $[\text{H}^+]$

Explanation (requires correct answer)

3. Which of the following best describes the mechanism by which hydrogen ions affect the affinity of hemoglobin for oxygen?

- A. Decrease the affinity by binding directly to oxygen binding sites
- B. Decrease the affinity by causing a conformational change in hemoglobin
- C. Have no effect on the affinity
- D. Increase the affinity by binding directly to oxygen binding sites
- E. Increase the affinity by causing a conformational change in hemoglobin

Explanation (requires correct answer)

4. Which of the following is the most abundant site of transport of CO_2 from the tissues to the lungs?

- A. Bicarbonate dissolved in RBCs
- B. CO_2 bound to hemoglobin
- C. Dissolved CO_2 in RBCs
- D. Dissolved CO_2 in plasma
- E. Plasma bicarbonate

Explanation (requires correct answer)

How would you rate this Brick?



Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- 01 Describe the mechanisms by which oxygen (O_2) and carbon dioxide (CO_2) are transported in the blood.
- 02 Explain the importance of the blood CO_2 concentration.
- 03 Relate the oxygen-hemoglobin dissociation curve to gas transport.
- 04 Apply the Haldane and Bohr effects to model oxygen and CO_2 uptake and release at the lung and peripheral tissues.

Objective Confidence: 0/4